Mira Chami

Applied AI Engineer

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Applied AI engineer building grounded generation, knowledge-constrained dialogue and controlled model actions inside interactive products. Five years of professional engineering across game engines, backend services, cross-platform desktop applications and shipped games.

PROFESSIONAL EXPERIENCE

Tencent

Nov 2023 – Present

Game Tech R&D Engineer · London, UK

Investigative Game & Applied AI

Gemini, Node.js, Godot

- Built a source-linked, **multi-stage Gemini pipeline** for source gathering, evidence extraction, witness research and narrative generation. Validated each stage's structured output against JSON Schemas; retained citations only from pages retrieved through Google Search grounding.
- Implemented AI witnesses with **explicit knowledge boundaries**, separating character knowledge, beliefs and player suggestions. Verified each reply against that knowledge, allowed one revision for rejected replies and retained exact quotes for source-attributed clues.
- Kept question generation separate from answering: questions use only player-selected evidence and are reviewed before submission. Owned the Godot case-file, evidence-board and interview UI.
- Enabled automated testing without live model calls through mocked responses, schema/data-integrity checks and UI tests across viewport sizes. Built versioned case packages and a comparison viewer, plus Node.js/Express services with session restoration, caching, validation, timeouts, cancellation and retries.

LLM Game Director & Designer Tools

Godot, GDScript

- Built a validated gameplay action interface for an LLM director to spawn enemies, grant item rewards and change progression through server-authoritative systems, with shared events and entity context as model input.
- Built a timeline runtime and Godot editor plugin that designers used to create behaviour timelines for the AI director. It let them author, preview and validate behaviours without code using the same typed actions as the AI director, while model integration developed.
- Architected server-authoritative co-op multiplayer supporting Steam P2P, LAN, listen and dedicated servers, plus data-driven runs providing the foundation the AI director acts on.

Desktop AI Companion

React, Rust, Tauri

- Built the React/Rust client for a game-aware AI companion: prompts with session and game context, HTTP/SSE response streaming, progressive rendering and persistent per-user chat history across full-chat and companion interfaces.
- Led the Electron-to-Tauri evaluation and migration to Tauri v2; built multi-window chat, companion, avatar and settings UI; automated Windows/macOS CI builds and packaging with diagnostic panels and structured logging.

Online Services & Partner Integration

UE5, C++, Blueprints

- Built a UE5 multiplayer demo to demonstrate and validate InGame, Tencent's public online services framework, covering authentication, lobbies, matchmaking, quests, inventory and achievements in gameplay.
- Implemented PlayStation 5 support and dedicated-server debugging; supported partner studios with issues across plugin, backend and UI layers; modified CI pipelines.

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PROFESSIONAL EXPERIENCE (CONTINUED)

The Video Games Studio (VGS)

Jun 2020 – Aug 2022

Game Programmer · Beirut, Lebanon

- **Shipped two Unity mobile games**, owning gameplay architecture, persistence and progression from prototype to release.
- Integrated PlayFab and third-party backend services, and built a separate tool to run live competition cycles and retrieve winner data.
- Enabled content changes without code through ScriptableObject-driven difficulty, progression and shop configuration.
- Led programming for a Unity adventure prototype, translating design specifications into character, ability and input systems.
- Supported technical interviews, mentored developers, ran code reviews and maintained gameplay and software architecture documentation.

TECHNICAL TOOLKIT

Applied AI:
LLM Integration · Context Management · LLM Tool Design · Google Search Grounding · Structured Outputs (JSON Schema) · LLM Output Validation · LLM Response Streaming · Source Attribution

Languages:
Python · TypeScript · JavaScript · C++ · C# · Rust · GDScript · Java · C

Backend & Networking:
Node.js · Express · HTTP · SSE · gRPC · Protobuf · InGame · PlayFab · Steamworks · Photon PUN 2

Engines & Desktop:
Unreal Engine 5 · Unity · Godot · Blueprints · React · Tauri · Electron

Build & Engineering:
Git · Perforce · CI · Automated Testing · Mocking · Build & Release Automation · Structured Logging · Performance Profiling

Graphics & Native APIs:
Three.js · PixiJS · Unity URP · Native API Integration

EDUCATION

MSc Games Development (Programming) · Kingston University London · 2022 – 2023

BSc Computer Science · American University of Beirut · 2017 – 2021

BA Economics · American University of Beirut · 2016 – 2020

MISCELLANEOUS

Languages: English, French and Arabic - native